

Project Report

Auto Ticket Classification using Flow Designer

Team ID: SWTID-2026-3677

1. INTRODUCTION

1.1 Project Overview

Service desks receive a steady stream of tickets that must be categorised and sent to the right support team. When agents do this by hand it is slow and inconsistent. This project builds an automated flow in ServiceNow Flow Designer that runs when a new ticket is created, reads its short description and description, matches them against classification conditions and keywords, and sets the category, priority and assignment group. A notification is then sent and a dashboard shows the classified tickets. The project was carried out by a team of five over two sprints.

1.2 Purpose

The purpose of the project is to remove manual ticket triage so that tickets reach the right team on the first attempt. The goals are to reduce misrouted tickets and reassignments, speed up resolution, help the team meet SLAs, and show how a no-code automation on ServiceNow can solve a real service desk problem.

2. IDEATION PHASE

2.1 Problem Statement

In ServiceNow service desks, incoming tickets are classified and routed manually by agents. This is slow and inconsistent, and it leads to misrouted tickets, repeated reassignments, delayed resolution and SLA breaches.

2.2 Empathy Map Canvas

The empathy map below is for the service desk agent, the main person affected by manual triage.

SAYS	THINKS
• "I spend too much time sorting tickets." • "This ticket is in the wrong queue."	• "There must be a faster way to route these." • "Will we miss the SLA again?"
DOES	FEELS
• Reads each ticket and chooses the category by hand • Reassigns tickets that were sent to the wrong team • Chases the backlog before SLA deadlines	• Overloaded by repetitive work • Frustrated by misrouted tickets • Under pressure from the growing backlog
PAINS	GAINS
• Manual effort and slow triage • Misrouting and reassignment • Delays and SLA breaches	• Tickets routed automatically • Faster first response • More time spent solving problems

2.3 Brainstorming

The team considered several ways to classify tickets automatically:

- Keyword and condition based classification using a Flow Designer flow.
- Machine-learning classification using ServiceNow Predictive Intelligence.
- Fixed assignment rules based on the ticket category chosen by the requester.
- Email-based routing to separate team mailboxes.
- A Virtual Agent that asks the requester questions and sets the category.

The first idea was selected because it is no-code, needs no training data, runs inside ServiceNow and can be changed easily by an administrator.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Stage	Raise ticket	Ticket created	Classification	Assignment	Resolution and update
User action	Requester reports an issue	Ticket is saved in ServiceNow	Flow reads the description	Ticket reaches the support team	Agent solves the ticket
Touchpoint	Service Portal, email, incident form	Incident table	Flow Designer	Assignment group	Email notification
Experience	Wants quick help	Waits for a response	Happens automatically	Right team first time	Informed of progress
How the solution helps	Easy ways to submit	Flow starts immediately	Category, priority and group set	No manual routing or reassignment	Faster resolution

3.2 Solution RequirementFunctional requirements

FR No.	Requirement	Description
FR-1	Ticket creation	Requesters can raise tickets through the Service Portal, email or the incident form.
FR-2	Flow trigger	The flow starts automatically when a new ticket is created.
FR-3	Ticket classification	The flow reads the short description and description and matches them to a category using conditions and keywords.
FR-4	Ticket update	The flow sets the category, priority and assignment group on the ticket.
FR-5	Notification	Email notifications are sent to the requester and the agent.
FR-6	Reporting	A dashboard shows the classified tickets.

Non-functional requirements

NFR No.	Requirement	Description
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NFR-1	Usability	Requesters need no extra steps; classification is automatic.
NFR-2	Reliability	Every new ticket runs through the flow, and tickets with no match are left for manual review.
NFR No.	Requirement	Description
NFR-3	Performance	Classification runs as soon as the ticket is created, without agent action.
NFR-4	Maintainability	Administrators can change categories and keywords in the flow without writing code.
NFR-5	Scalability	More categories, ticket types and departments can be added by editing the rules.
NFR-6	Security	Access is controlled by ServiceNow user roles and access control rules.

3.3 Data Flow Diagram

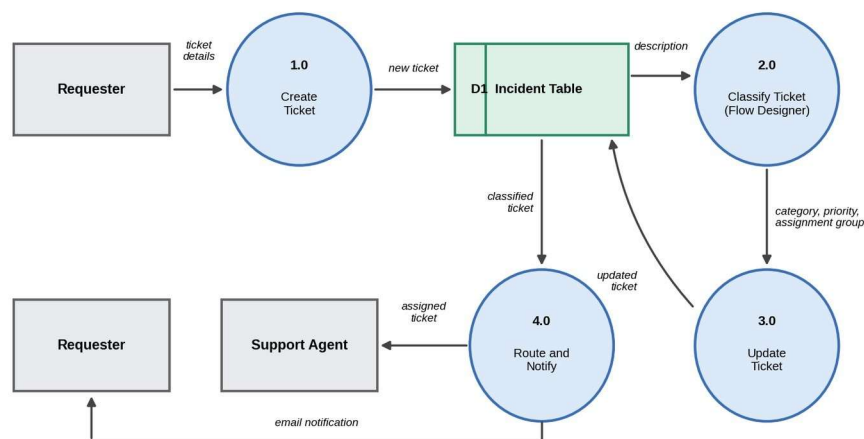


Figure 1: Data flow diagram of the ticket classification flow

3.4 Technology Stack

Component	Technology
Platform	ServiceNow (Personal Developer Instance)
Automation	ServiceNow Flow Designer
Data storage	ServiceNow Incident table
User interface	Service Portal and incident form
Notifications	ServiceNow email notifications
Reporting	ServiceNow dashboard
Access control	User roles and access control rules

4. PROJECT DESIGN

4.1 Problem Solution Fit

The Problem-Solution Fit canvas below shows the customer segments, their problems, triggers, constraints and behaviours, and how the Flow Designer solution fits them.



Figure 2: Problem-Solution Fit canvas

4.2 Proposed Solution

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	In ServiceNow service desks, incoming tickets are classified and routed manually by agents. This is slow and inconsistent, and it leads to misrouted tickets, repeated reassignments, delayed resolution and SLA breaches.
2.	Idea / Solution description	An automated flow built in ServiceNow Flow Designer that triggers when a new ticket is created. It reads the ticket's short description and description, identifies the category using conditions and keywords, then sets the category, priority and assignment group and updates the ticket automatically.
3.	Novelty / Uniqueness	Uses only the platform's native low-code Flow Designer, with no external tools, custom scripts or model training. The classification rules are easy to read, change and extend by administrators, and everything runs inside ServiceNow.

4.	Social Impact / Customer Satisfaction	Users get faster responses because tickets reach the right team first time. Fewer misrouted tickets means quicker resolution, and agents spend less time on repetitive triage and more on solving problems, which improves satisfaction for both users and support staff.
5.	Business Model (Revenue Model)	Offered as a ServiceNow implementation service for organisations using the platform: a one-time setup and configuration fee, plus an optional paid support and maintenance plan for updating rules and adding new categories.
S.No.	Parameter	Description
6.	Scalability of the Solution	The flow can be extended to more categories, ticket types (incidents, requests, changes) and departments by adding or editing rules. Because it runs natively on ServiceNow, it can handle growing ticket volumes without additional infrastructure.

4.3 Solution Architecture

A requester raises a ticket through the Service Portal, email or the incident form. The ticket is stored in the Incident table, which triggers the Flow Designer flow. The flow classifies the ticket and updates its category, priority and assignment group. The ticket is then routed to the assignment group, the support agent resolves it, and a notification is sent.

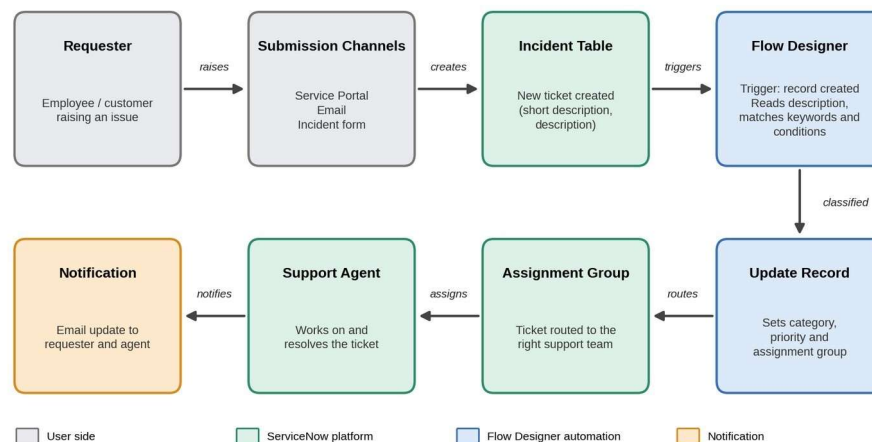


Figure 3: Solution architecture and data flow

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The work was divided into five epics and eleven user stories across two sprints.

Sprint	USN	Epic	User Story / Task	Points	Priority
Sprint-1	USN-1	ServiceNow Setup	Set up a ServiceNow developer instance to build the ticket classification solution.	2	High
Sprint-1	USN-2		Create categories and assignment groups so that tickets can be classified and routed.	1	High

Sprint-1	USN-3	Ticket Data Preparation	Define classification rules and keywords for each ticket category.	3	High
Sprint-1	USN-4		Create custom fields for category and priority on the incident form.	3	Medium
Sprint-1	USN-5		Handle inconsistent ticket descriptions so that classification works reliably.	3	Medium
Sprint-2	USN-6	Flow Designer Automation	Create a flow trigger that runs when a new ticket is created.	2	High
Sprint	USN	Epic	User Story / Task	Points	Priority
Sprint-2	USN-7		Make the flow read the ticket's short description and description.	2	High
Sprint-2	USN-8		Apply classification conditions to match each ticket to a category.	2	High
Sprint-2	USN-9		Update the category, priority and assignment group automatically.	4	High
Sprint-2	USN-10	Testing	Test the flow with sample tickets to confirm they are classified correctly.	5	High
Sprint-2	USN-11	Notification and Reporting	Send email notifications and show a dashboard of classified tickets.	5	Medium

Sprint schedule and velocity

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	12	6 Days	05 Oct 2026	10 Oct 2026
Sprint-2	20	6 Days	12 Oct 2026	17 Oct 2026

Total story points = 12 + 20 = 32 over 2 sprints, so the team velocity is $32 / 2 = 16$ story points per sprint. With a 6-day sprint, the average velocity is $16 / 6 \approx 2.67$ story points per day.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The flow was tested by creating sample tickets with different descriptions and checking in Flow Designer's execution details that the flow ran and that each ticket received the expected category and assignment group. The flow starts as soon as a ticket is created, so classification needs no action from an agent. The test scenarios are shown below.

Test	Sample ticket description	Expected result
T1	Cannot connect to VPN from home	Category Network; assigned to Network Support

T2	Outlook crashes when opening attachments	Category Software; assigned to Application Support
T3	Laptop screen is flickering	Category Hardware; assigned to Hardware Support
T4	Forgot my password and account is locked	Category Access; assigned to Access Management
T5	Need help (no matching keyword)	Not classified; left for manual review

7. RESULTS

7.1 Output Screenshots

servicenow All Favorites History Incident WorkFlow - Create INC000... Search

Incident WorkFlow
New record

Submit

Number

Caller

Category

Subcategory

Short Description

Description

State

Assigned group

Assigned to

Submit

Incident WorkFlow
INC0001003

Update Delete

Number

Caller

Category

Subcategory

Short Description

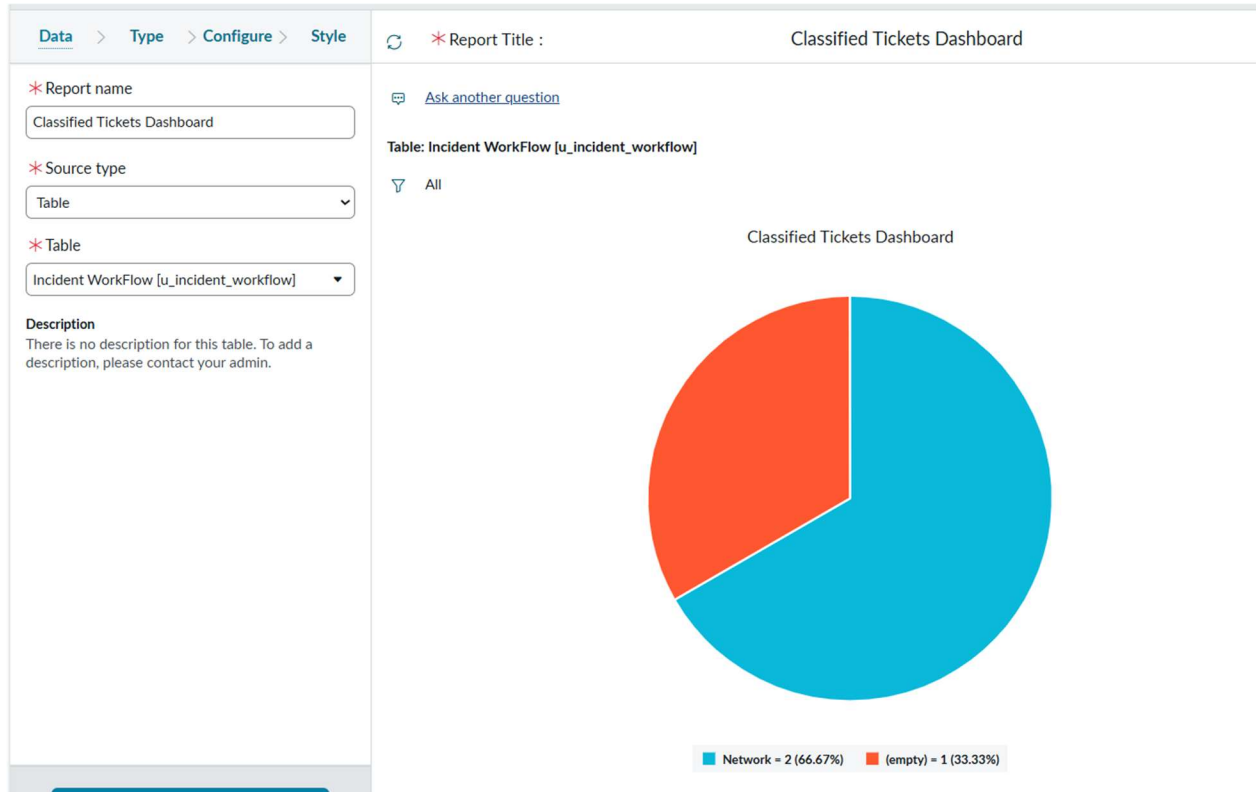
Description

State

Assigned group

Assigned to

Update Delete



8. ADVANTAGES & DISADVANTAGES

Advantages

- Removes manual triage, so agents spend more time solving problems.
- Tickets reach the right team first time, reducing reassignments and delays.
- Built with a no-code tool, so rules are easy to read, change and extend.
- Runs inside ServiceNow, with no extra infrastructure or external tools.

Disadvantages

- Depends on the conditions and keywords defined, so vague tickets may not be classified correctly.
- New categories and keywords must be added by an administrator.
- Needs a ServiceNow instance and Flow Designer access.

9. CONCLUSION

The project shows that ticket classification and routing can be automated on ServiceNow using Flow Designer alone. By setting the category, priority and assignment group as soon as a ticket is created, the solution reduces manual effort, avoids misrouting and helps tickets get resolved faster. The same approach can be extended to more categories and ticket types as the service desk grows.

10. FUTURE SCOPE

- Use machine-learning classification (for example ServiceNow Predictive Intelligence) alongside the keyword rules.
- Add more categories, ticket types (requests, changes) and departments.
- Add automatic SLA tracking and a Virtual Agent for ticket creation.

11. APPENDIX

- **Source Code (if any):** Not applicable. The solution was built with Flow Designer (lowcode), so no custom source code was written. The flow can be exported as an update set.
- **Dataset Link:** Not applicable. Sample tickets were created manually for testing.
- **GitHub & Project Demo Link:** [Link to be added]